



IBM watsonx.ai platform efficient usage tips

IBM watsonx.ai platform consumes resources by tokens for foundational model tools like the Prompt Lab and by computing unit hours (CUH) for machine learning tools/services like the Watson Studio and the Watson Machine Learning.

Tokens

A token is a collection of characters that has semantic meaning for a model. Tokenization is the process of converting the words in your prompt into tokens. You can learn more about tokens and how they are calculated [here](#).

CUH

Many tools consume compute usage that is measured in capacity unit hours (CUH). A capacity unit hour is a specific amount of compute capability with a set cost. You can learn more about CUH and how it is calculated [here](#).

You can monitor the token and CUH usage in a project on the **Environments** page on the **Resource usage** tab.

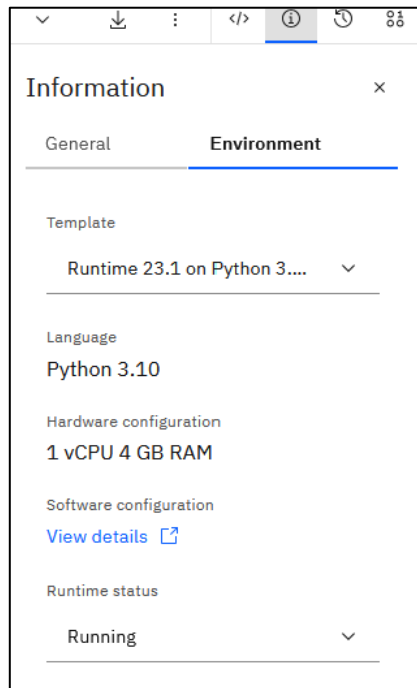
Tips to work efficiently on the watsonx.ai platform

Prompting:

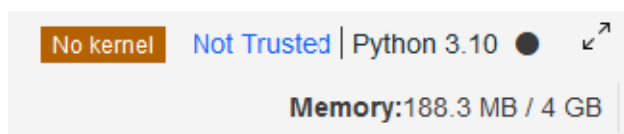
- Write clear, concise, and specific prompts to minimize the need for revisions and reduce token consumption.
- Leverage Prompt Lab's model parameters to set limits for token consumption and repetitions for efficient prompting. Monitor your usage and adjust your prompts accordingly to stay within your limits.
- While working with API endpoints, limit the frequency of calling the endpoint only when required within your application.

Jupyter Notebook:

- Select only the required runtime environment to run your notebook to cut down the CUH consumption.
- After successful notebook execution, be sure to stop the kernel and monitor the memory usage at the top-right of the notebook so CUH consumption is not running in the background.
- You can stop the kernel and modify your runtime environment by selecting the information icon at the top-right of the notebook. You can always start the kernel to run the notebook.



- If the kernel has been stopped, a “No kernel” notification will be displayed, and the Python runtime will be displayed as a black dot.



- If the kernel is running, the Python runtime will be displayed as a transparent dot. Notice the increase in the memory consumption.



- Whenever you stop the kernel runtime, your work will be saved automatically, and you can always access and edit the saved notebook from the project’s **Assets** tab.

General:

- Collaborate with other participants to form a team and leverage your team members’ cloud accounts to utilize the resources effectively.
- Participants are not allowed to use the provisioned hackathon IBM Cloud account services for any purposes other than to build a solution for the hackathon.
- Always save your work in regular intervals. Learn about saving your watsonx.ai work [here](#).